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Lorikeet Science Unit 3&4 Chemistry Trial Exam 2025

Chemistry (MacRobertson Girls High School)



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TRIAL CHEMISTRY EXAM

Name _____

Unit 3 & 4 Chemistry

Question and Answer Book

VCE Trial Examination

- Reading time is **15 minutes**
- Writing time is **2 hours 30 minutes**

Approved materials

- One scientific calculator

Materials supplied

- Question and Answer Book of 39 pages
- Data Book
- Multiple-Choice Answer Sheet

Instructions

- Follow the instructions on your Multiple-Choice Answer Sheet
- At the end of the examination, place your Multiple-Choice Answer Sheet inside the front cover of this book.

Students are not permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room

Contents pages

Section A (30 questions, 30 marks) _____ 2 - 17

Section B (12 questions, 90 marks) _____ 18 - 39

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Section A - Multiple-choice questions

Instructions

- Answer all questions in pencil on the Multiple-Choice Answer Sheet.
- Choose the response that is correct or that best answers the question.
- A correct answer scores 1; an incorrect answer scores 0.
- Marks will not be deducted for incorrect answers.
- No marks will be given if more than one answer is completed for any question.
- Unless otherwise indicated, the diagrams in this book are not drawn to scale.

Question 1.

Identify the option containing only biofuels

A. biodiesel, coal, bioethanol

B. biogas, natural gas, biodiesel

C. petrol, bioethanol, biodiesel

D. bioethanol, biodiesel, biogas

Question 2.

Identify the correctly balanced equation that represents photosynthesis in living organisms.

A. $2CO_2(g) + 2H_2O(l) \rightarrow C_6H_{12}O_6(aq) + 2O_2(g)$

B. $C_6H_{12}O_6(aq) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$

C. $6CO_2(g) + 6H_2O(l) \rightarrow C_6H_{12}O_6(aq) + 6O_2(g)$

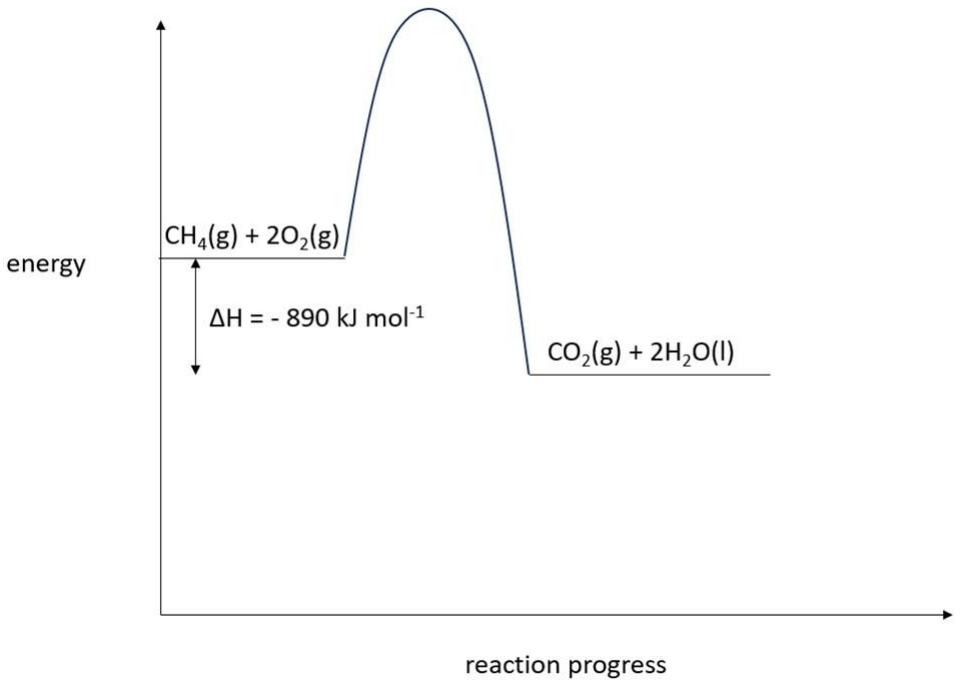
D. $C_6H_{12}O_6(aq) + CO_2(g) \rightarrow O_2(g) + H_2O(l)$

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Question 3.



Identify the correct statement

A. the reaction is endothermic and the molar enthalpy change of methane is 890 kJ mol^{-1}

B. 2 moles of methane when combusted would release 890 kJ of energy

C. methane and oxygen are more stable than carbon dioxide and water

D. energy is required to break bonds in the reactants

Question 4.

50.0 g of butane gas is completely combusted in oxygen at SLC.

The equation is $2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(l)$

Calculate the volume, in L, of CO_2 produced. Assume that oxygen gas is in excess.

A. 0.862

B. 3.45

C. 85.5

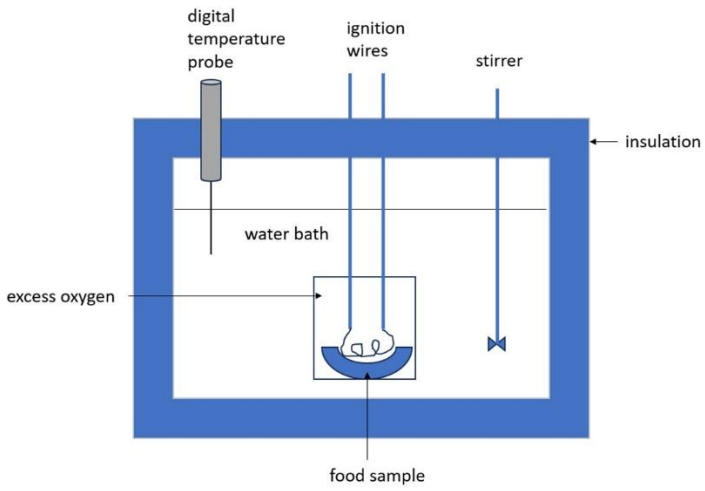
D. 342

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Question 5.

A 2.50 g sample of dry pasta was placed in a bomb calorimeter.



The initial temperature of the water in the calorimeter was 20.2 ºC, after complete combustion the final temperature of the water was 27.9 ºC. The calibration factor for the calorimeter was 5.54 kJ $ºC^{-1}$. Calculate the energy released, in kJ per g, of dry pasta

A. 5.5

B. 7.7

C. 17

D. 43

Question 6.

Deduce the equation that represents a redox reaction.

A. $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

B. $HCOOH(aq) + NaOH(aq) \rightarrow NaCOO(aq) + H_2O(l)$

C. $Hg(NO_3)_2(aq) + 2KI(aq) \rightarrow 2KNO_3(aq) + HgI_2(s)$

D. $H_2SO_4(aq) + 2LiOH(aq) \rightarrow Li_2SO_4(aq) + 2H_2O(l)$

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Question 7.

Consider the following redox equation.

$Cr_2O_7^{2-}(aq) + 8H^+(aq) + 3H_2O_2(aq) \rightarrow 2Cr^{3+}(aq) + 3O_2(g) + 7H_2O(l)$

Identify the oxidising agent.

A. Cr

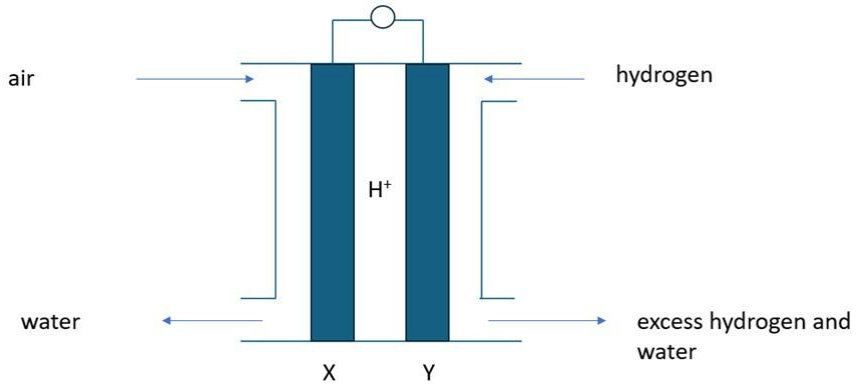
B. H^+

C. H_2O_2

D. $Cr_2O_7^{2-}$

Question 8.

Consider the fuel cell represented in the following diagram:



Deduce the option that correctly identifies electrode Y and the process occurring here.

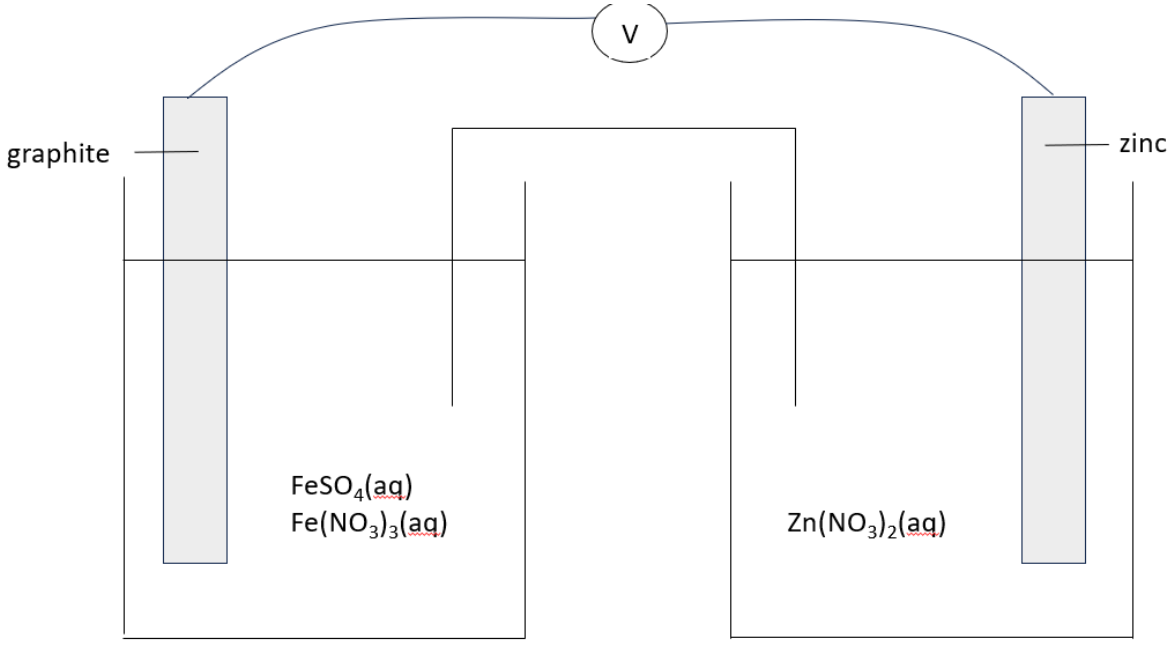
	Electrode Y	**Process**
A.	Anode	Reduction
B.	Cathode	Reduction
C.	Anode	Oxidation
D.	Cathode	Oxidation

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Question 9.

A galvanic cell is represented by the following diagram.



Identify the cathode and half-reaction that takes place at the cathode when the circuit is connected.

	cathode	**reaction at cathode**
A.	Zinc metal	$Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$
B.	Zinc metal	$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$
C.	Graphite	$Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$
D.	Graphite	$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$

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Question 10.

Consider the following reaction

$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$ $K = 8.3 \times 10^6\ M^{-1}$ at 450 ^oC

Identify the correct statement.

A. K is very large and at equilibrium there is a significantly higher proportion of reactants than products

B. K is very large and at equilibrium there is a significantly higher proportion of products than reactants

C. K is very small and at equilibrium there is a significantly higher proportion of reactants than products

D. K is very small and at equilibrium there is a significantly higher proportion of products than reactants

Question 11.

Consider the gaseous equilibrium system

$$N_2O_4(g) \rightleftharpoons 2NO_2(g) \qquad \Delta H>0$$

colourless brown

Predict the outcome if the temperature of the equilibrium system is increased

A. The equilibrium mixture would gradually become darker brown

B. The colour of the equilibrium mixture would not change

C. The equilibrium mixture would gradually become lighter brown

D. The equilibrium mixture would immediately become lighter brown then darker in colour but not as dark as the initial colour

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Question 12.

Consider the gaseous equilibrium system

$Br_2(g) + I_2(g) \rightleftharpoons 2IBr(g)$ K = 1.2 x 10^2 at 150 ^oC

Deduce the equilibrium constant, K for the reaction $3Br_2(g) + 3I_2(g) \rightleftharpoons 6IBr(g)$ at 150 ^oC

A. 8.3 x 10^{-3}

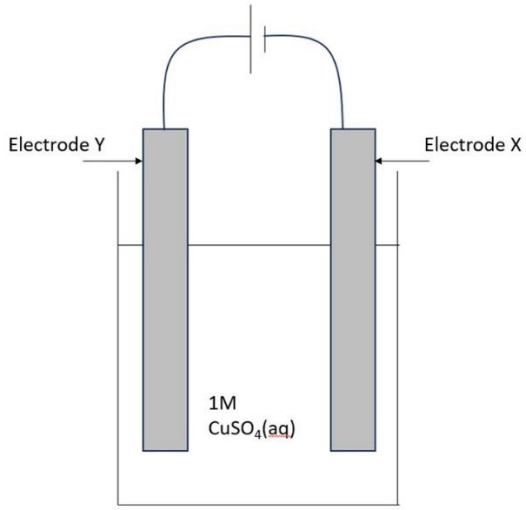
B. 1.4 x 10^4

C. 3.6 x 10^2

D. 1.7 x 10^6

Question 13.

An electrochemical cell is shown in the following diagram



Identify the correct alternative for the electrochemical cell.

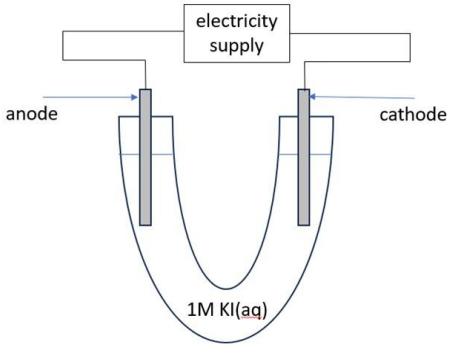
	Type of cell	**Polarity of Electrode X**
A.	electrolytic	positive
B.	fuel	positive
C.	electrolytic	negative
D.	fuel	negative

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Question 14.

A student electrolysed a 1M solution of sodium iodide, NaI, with graphite electrodes.



Predict a product formed at the anode.

A. $O_2(g)$

B. Na(s)

C. $I_2(aq)$

D. $H_2(g)$

Question 15.

Identify the degree of unsaturation of the molecule with semi-structural formula $CH_3CH_2CHCHCH_2CH_2CHCHCH_2CH_3$.

A. 0

B. 1

C. 2

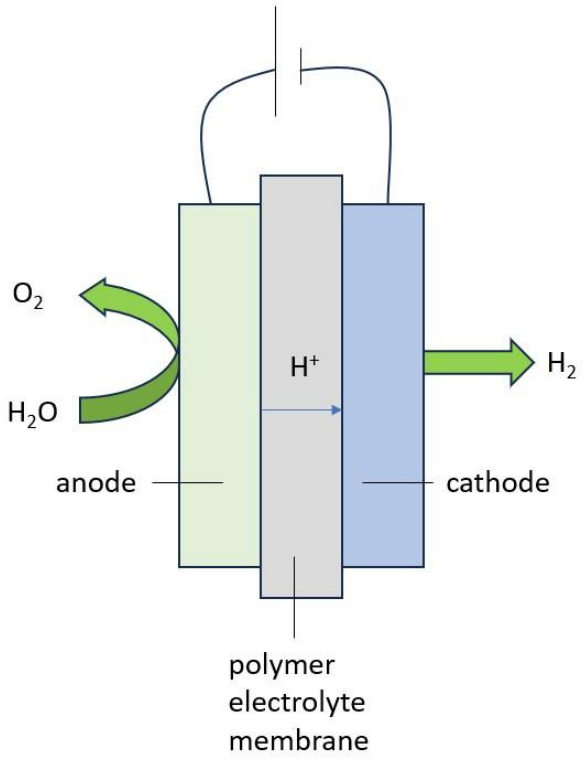
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Question 16.

Consider a Polymer Electrolyte Membrane (PEM) electrolysis cell to produce green hydrogen.



Deduce the half-equations that occur in this cell and the overall equation when operating

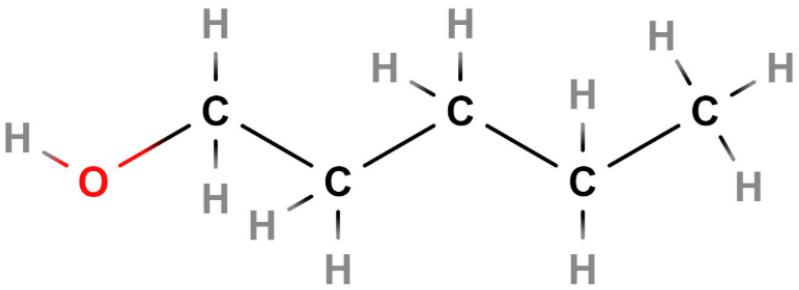
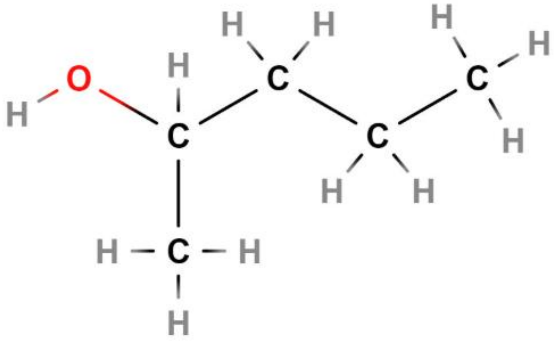
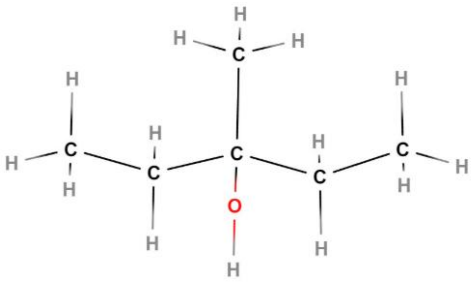
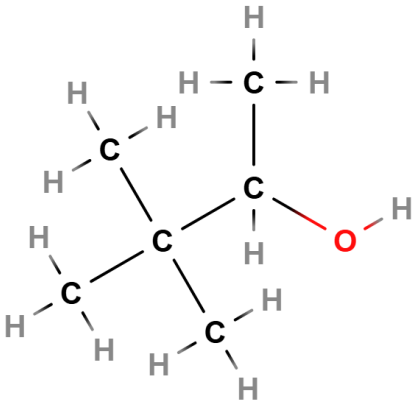
	Anode	**Cathode**	**Overall**
A.	$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$	$2H_2O(l) \rightarrow 4H^+ + O_2(g) + 4e^-$	$2H^+ + 2e^- \rightarrow H_2(g)$
B.	$2H_2O(l) \rightarrow 4H^+ + O_2(g) + 4e^-$	$2H^+ + 2e^- \rightarrow H_2(g)$	$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$
C.	$2H^+ + 2e^- \rightarrow H_2(g)$	$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$	$2H_2O(l) \rightarrow 4H^+ + O_2(g) + 4e^-$
D.	$2H^+ + 2e^- \rightarrow H_2(g)\|$	$2H_2O(l) \rightarrow 4H^+ + O_2(g) + 4e^-$	$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$

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Question 17.

Identify the structural formula that represents a tertiary alcohol

A.	
B.	
C.	
D.	

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Question 18.

Deduce the condensed (semi-structural formulas) that represents a primary amide

A.	$CH_3CH_2CH_2CONH_2$
B.	$CH_3CH_2CH_2CH_2NH_2$
C.	$CH_3CH_2CHCHCH_3$
D.	$CH_3CH_2CH_2CONHCH_3$

Question 19.

Hydrogen can be synthesized from methane using the steam reforming process according to the equation

$CH_4(g) + H_2O(g) \rightarrow CO(g) + 3H_2(g)$.

The next step involves oxidizing carbon monoxide to carbon dioxide.

The CSIRO classifies the hydrogen produced through this process as grey hydrogen because the carbon dioxide emissions are released to the environment.

Carbon monoxide is a highly toxic gas.

Identify the limiting reactant and calculate the mass of carbon monoxide that would be produced from the reaction of 1.00 kg methane and 1.00 kg steam.

	Name of limiting reactant	**Mass of CO produced (kg)**
A.	methane	55.6
B.	methane	62.5
C.	steam	1.56
D.	steam	1.75

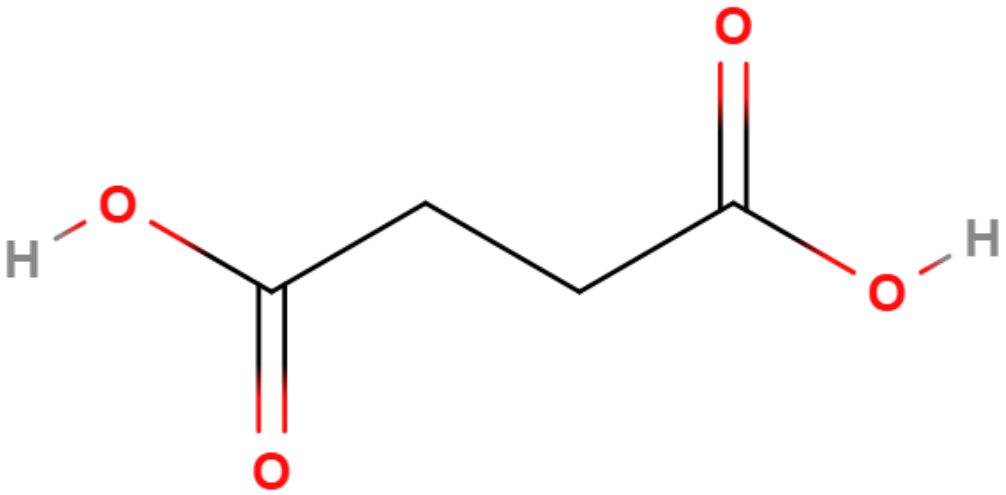
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Question 20.

Name the following molecule using IUPAC systematic nomenclature



A. 1,4-butanedioic acid

B. 1,4-dicarboxybutane

C. 1,4-butanediol

D. butanedioic acid

Question 21.

Identify the correct order of increasing boiling point of the alkanes: octane, methane, butane, hexane and pentane

A. methane < pentane < butane < octane < hexane

B. octane < pentane < hexane < butane < methane

C. methane < butane < pentane < hexane < octane

D. octane < hexane < pentane < butane < methane

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Question 22.

Identify the strongest type of intermolecular force that exists between adjacent molecules of biodiesel.

A. Covalent bonding

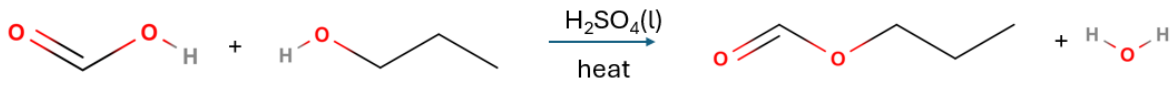
B. Hydrogen bonding

C. Dispersion forces

D. Dipole-dipole attraction

Question 23.

Deduce the following reaction type



A. esterification / hydration

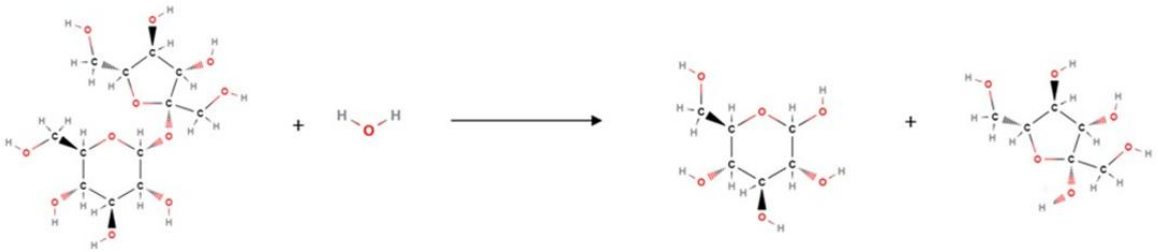
B. substitution / hydration

C. hydrolysis / addition

D. esterification / condensation

Question 24.

Identify the following reaction:



A. condensation to produce a disaccharide

B. hydrolysis of a polysaccharide

C. hydrolysis of a disaccharide

D. hydrolysis of a protein

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Question 25.

A high performance liquid chromatography (HPLC) instrument is set up with a polar mobile phase and a non-polar stationary phase. A mixture of three amino acids serine (Ser), threonine (Thr) and isoleucine (Ile) are added to the mobile phase and are run through the instrument.

Deduce the order of retention times, from shortest to longest for these three amino acids

A. isoleucine, threonine, serine

B. serine, isoleucine, threonine

C. threonine, isoleucine, serine

D. serine, threonine, isoleucine

Question 26.

State the representation the strongest covalent bond

A. C-H

B. C-C

C. C=C

D. C ≡ C

Question 27.

Identify the type of reaction that occurs when methanol is reacted with triglycerides in the presence of a catalyst.

A. esterification

B. transesterification

C. addition

D. substitution

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Question 28.

Recall the information supplied by the integration curve (area under the peak) in ^{1}H NMR spectroscopy

A. the type of chemical environment

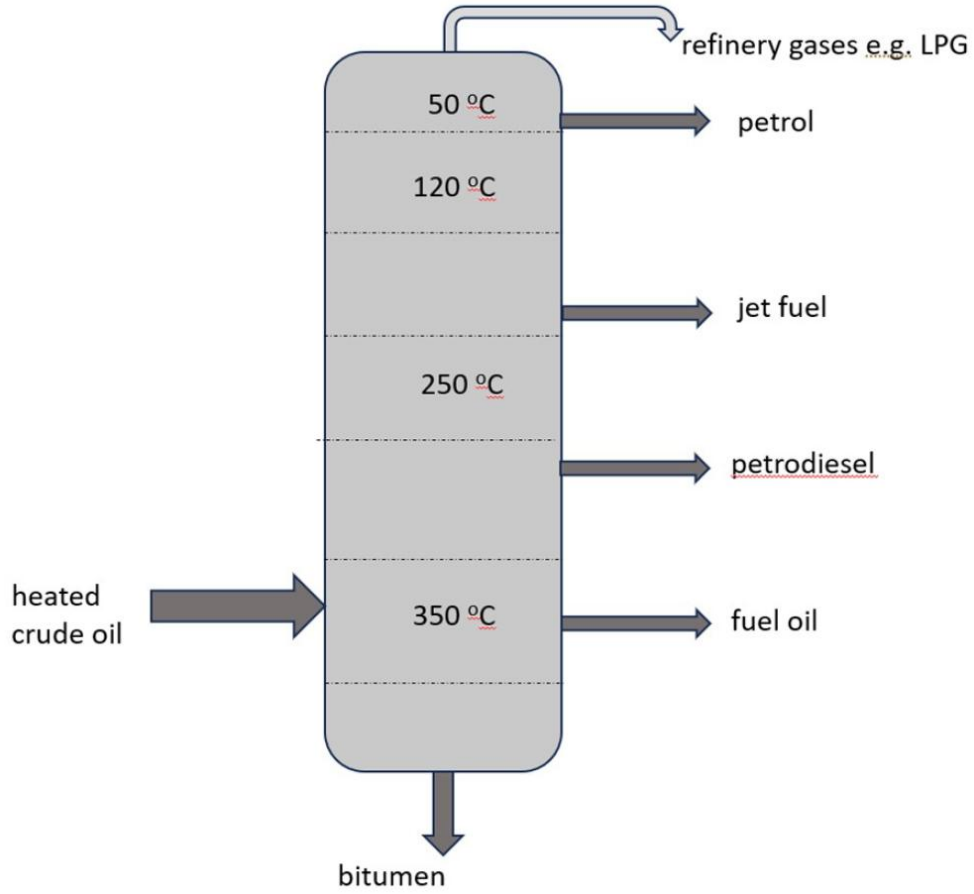
B. the number of hydrogens in a particular environment

C. the ratio of the number of hydrogens in a particular environment

D. the number of unique hydrogen environments

Question 29.

Consider the following diagram.



Name this technique

A. Melting point determination

B. Solvent extraction

C. Simple distillation

D. Fractional distillation

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Question 30.

Identify the type(s) of bonding is/are involved in the primary structure of enzymes

A. Hydrogen bonding only

B. Covalent bonding in amide bonds only

C. Hydrogen bonding and ionic bonding only

D. Ionic bonding, hydrogen bonding, dispersion forces and covalent bonding in disulfide bridges

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Section B

Instructions

- Answer all questions in the spaces provided.
- Write your responses in English.
- Give simplified answers to all numerical questions, with an appropriate number of significant figures; unsimplified answers will not be given full marks.
- Show all working in your answers to numerical questions; no marks will be given for an incorrect answer unless it is accompanied by details of the working.
- Ensure that chemical equations are balanced and that the formulas for individual substances include an indication of state, for example, $H_2(g)$, NaCl(s). Unless otherwise indicated, the diagrams in this book are not drawn to scale.

Question 1 (7 marks)

The Australian Mushroom Council is concerned about recent deaths of people who unknowingly consumed a meal containing poisonous Death Cap mushrooms. Death Cap mushrooms contain a highly toxic compound alpha-amantin which causes cell death and liver failure.

However, mushrooms sold in supermarkets and green-grocers are safe to consume because they do not contain the toxic compound.

a. Name a method that could be used to determine the presence of alpha-amantin in a mushroom sample 1 mark

b. Describe how this method could be used to identify whether alpha-amantin is present in the mushroom sample 3 marks

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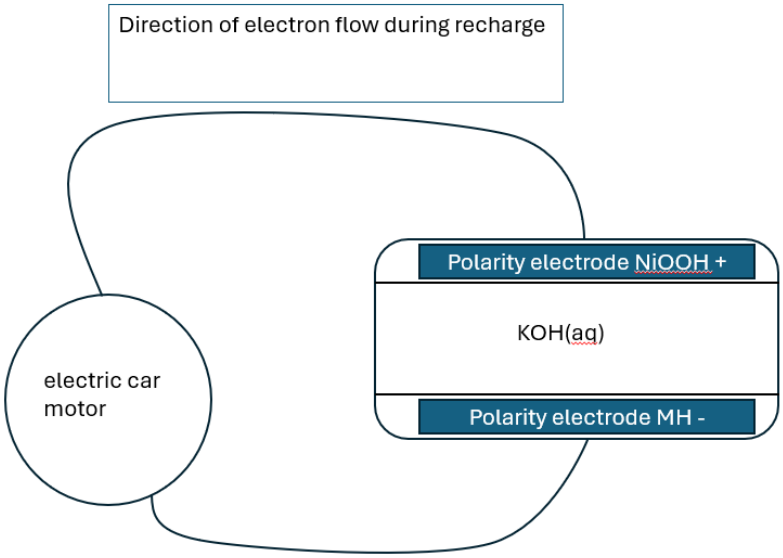
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c. Describe how a chemist could determine the concentration of alpha-amantin present in a sample found to contain this toxin. 3 marks

The results of this test can be used to reassure the public that consumption of regular mushrooms is safe.

Question 2 (7 marks)

A storage battery used in hybrid cars is comprised of a series of nickel metal hydride, NiMH, cells. MH represents a metal hydride alloy that is used as one electrode. The other electrode contains nickel oxide hydroxide, NiOOH. The electrolyte is aqueous KOH.



The equation representing the reaction occurring at the positive electrode during discharge is

$$NiOOH(s) + H_2O(l) + e^- \rightarrow Ni(OH)_2(s) + OH^-(aq)$$

The equation representing the reaction occurring at the negative electrode during discharge is

$$MH(s) + OH^-(aq) \rightarrow M(s) + H_2O(l) + e^-$$

When discharging at standard laboratory conditions the voltage was recorded as 1.25 V.

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a. Circle the option that best describes the cell represented 1 mark

primary **secondary** **fuel** **electrolytic**

b. Circle the process that occurs during discharge at the positive electrode 1 mark

oxidation **reduction**

c. Write the reactions at the anode and cathode when the cell is recharging 2 marks

anode	
cathode	

d. Write the overall reaction when the cell is recharging 1 mark

e. Annotate the cell diagram to show the direction of electron flow when the cell is recharging 1 mark

f. State the applied voltage required to recharge the cell 1 mark

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Question 3 (5 marks)

a. Sketch an electrolytic cell that has a nickel anode, a $NiSO_4$ electrolyte and contains a key to be fully electroplated with nickel 2 marks

b. Calculate the mass of nickel electroplated onto the key if the cell has a current of 5.00 A for 6.00 minutes. 3 marks

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Question 4 (13 marks)

a. Beef contains the amino acid alanine. State the IUPAC systematic name of alanine.

1 mark

The side chain on amino acids can be classified as polar, non-polar, acidic or basic.

b. Circle the type of side chain found in alanine 1 mark

polar **non-polar** **acidic** **basic**

c. Draw the structural formula of the tripeptide Glu-Ser-Asp at pH 2 3 marks

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d. A nutrition label on a packet of beef rump steak contains the following information. When cooked, a 180g serve of grilled rump steak contains

Protein 29.9 per 100g

Fat 10.9 per 100g

Carbohydrate 0g per 100g

Calculate the energy per 180 g serve 2 marks

e. The Estimated average requirement of protein for a 14-18 year old female adolescent living in Australia is 35 g per day. Calculate the mass of grilled steak that a female teenager would need to eat to obtain their daily protein intake? 1 mark

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f. Complete the reaction pathway for producing the amino acid, alanine, from propane.

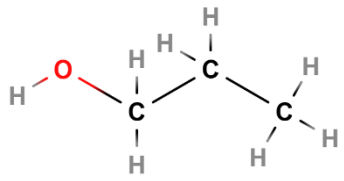
5 marks

Box Q

$Cl_2(g)$, UV light

Box R

Box S

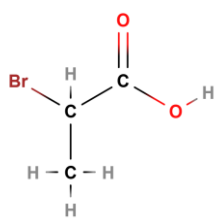


Box T MnO_4^-/H^+ or $Cr_2O_7^{2-}/H^+$

Box U

$Br_2(g)$, UV light

Box V 2-bromopropanoic acid



NH_3

alanine

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a. In Box Q draw the semi-structural formula of propane

b. In Box R draw the skeletal structure of the molecule formed

c. In Box S state the formula of the required reagent

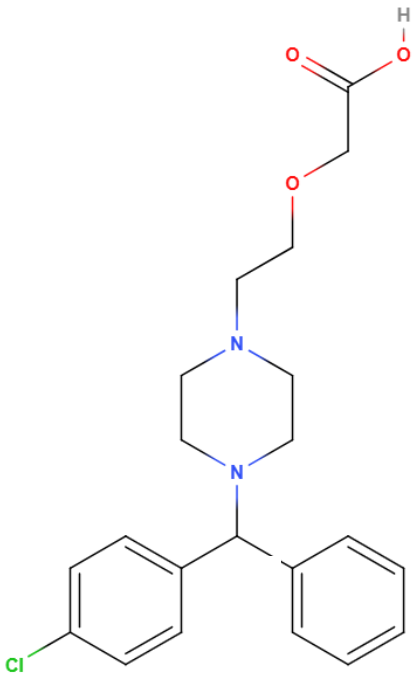
d. In Box T name the type of reaction taking place

e. In Box U name the molecule formed

Question 5 (5 marks)

Antihistamines can be taken orally to counter the effect of many symptoms of allergies.

Cetirizine is a type of antihistamine medication. Its skeletal structure can be represented by



a. Using an asterisk identify a chiral carbon in this molecule**.** 1 mark

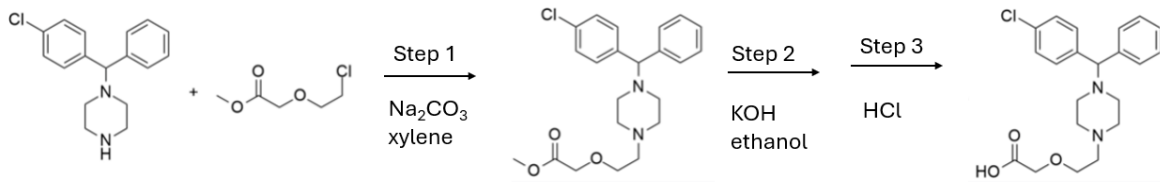
b. Circle and name two functional groups in this molecule. 1 mark

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c. Describe a qualitative test, together with the expected result, that could be conducted to identify the structure of one of the functional groups in the molecule 2 marks

Consider the following pathway.



Step 1 28% yield, Step 2 56% yield, Step 3, 81% yield.

d. Calculate the percentage yield when cetirizine is synthesised from chlorophenylmethyl-piperazine in the pathway outlined above. 1 mark

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Question 6. (7 marks)

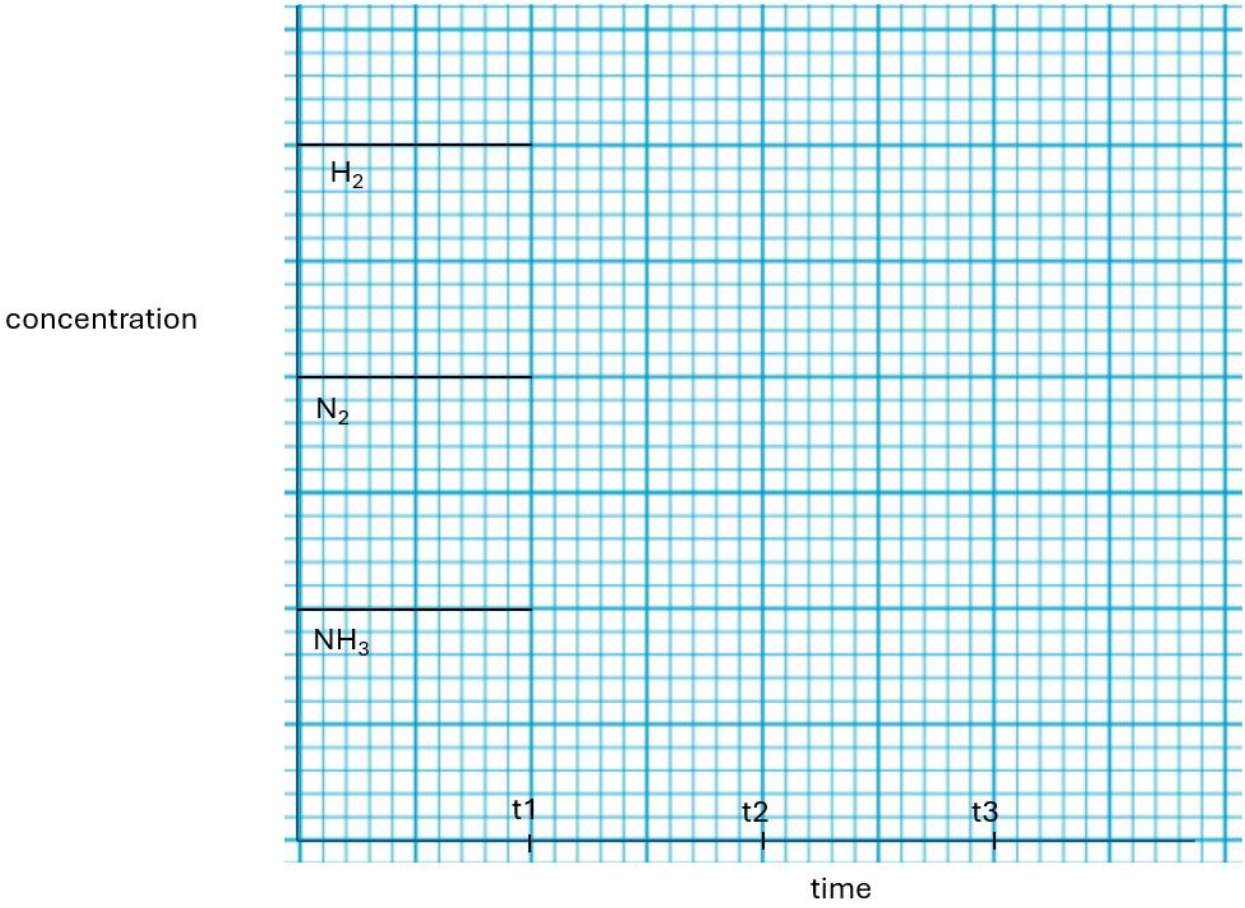
Consider the Haber process represented by the equilibrium system

$3H_2(g) + N_2(g) \rightleftharpoons 2NH_3(g)$ $\Delta H < 0$

a. State the equilibrium expression for this reaction 1 mark

---

b. Draw the continuation of the graphs of the concentration of each species at



i. t1 when the volume of the system was doubled. 2 marks

ii. t2 when hydrogen gas was added to the system 2 marks

iii. t3 when the temperature of the system was decreased 2 marks

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Question 7 (7 marks)

a. Write the equation, using molecular formulas, for the fermentation of glucose to produce ethanol 1 mark

b. Name the technique that is used after fermentation to purify the ethanol 1 mark

c. Explain the principle behind this technique that results in the purification of the ethanol. 1 mark

d. Write the equation for the complete combustion of pure ethanol 1 mark

e. Using semi-structural formulas write an equation to represent the reaction of ethanol with propanoic acid. Include all conditions required. 1 mark

f. Name the desired product formed in part e. 1 mark

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g. Calculate the percentage atom economy for the desired product in reaction e.

1 mark

Question 8 (3 marks)

In July 2025, Food Standards Australia New Zealand (FSANZ) put out a notification that the enzyme phospholipase c be permitted as a processing aid in the processing of fats and oils.

https://mail.google.com/mail/u/0/?tab=rm&ogbl#inbox/FMfcgzQbgJNQJfQzXZqjFrKchXfWfKlW

a. Name the model that describes the process by which an enzyme catalyses a biological reaction.

1 mark

---

b. Complete the table by identifying a variable that must be controlled to avoid the denaturation of an enzyme. Outline the specific interactions important in the functioning of the enzyme that may be impacted if this variable has not been controlled.

2 marks

Variable	**Explanation**

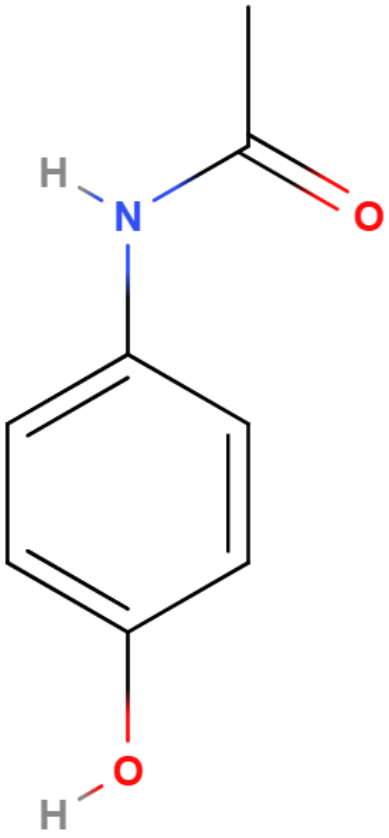
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Question 9 (8 marks)

Paracetamol, $C_8H_9NO_2$, is a painkilling medication. It can be represented as

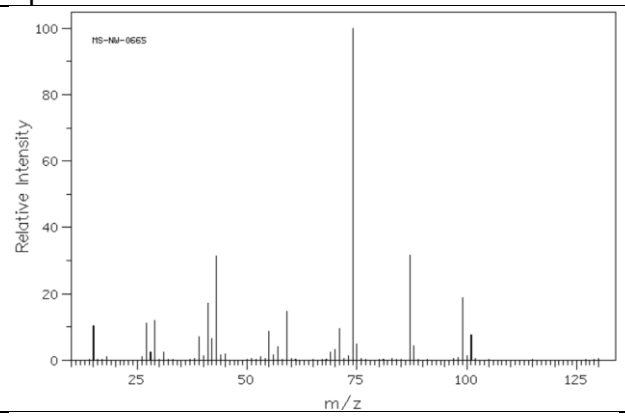
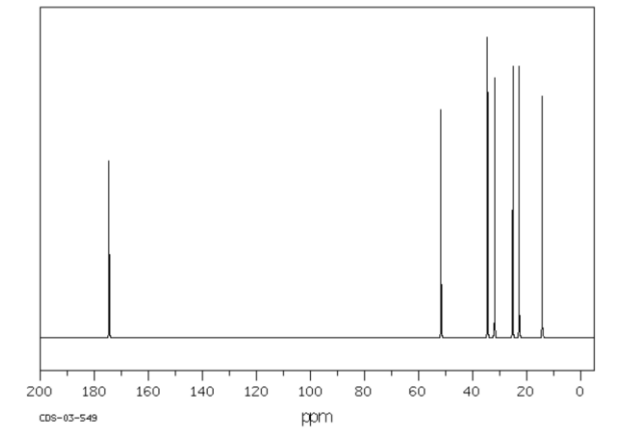
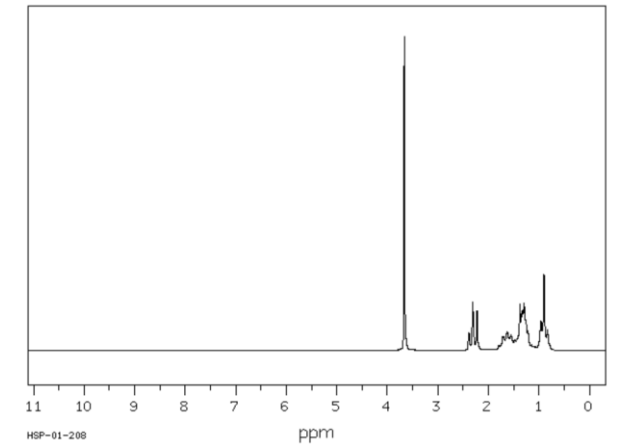
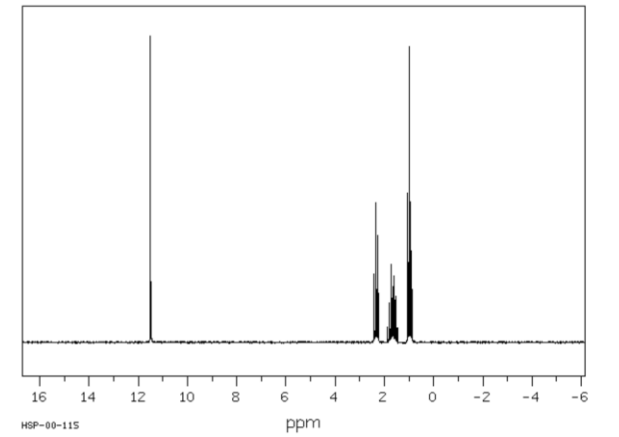


Consider spectra coded A, B, C, D, E, F, G, H, I, J, K, L Data: SDBSWeb; http://sdbs.db.aist.go.jp

(National Institute of Advanced Industrial Science and Technology)

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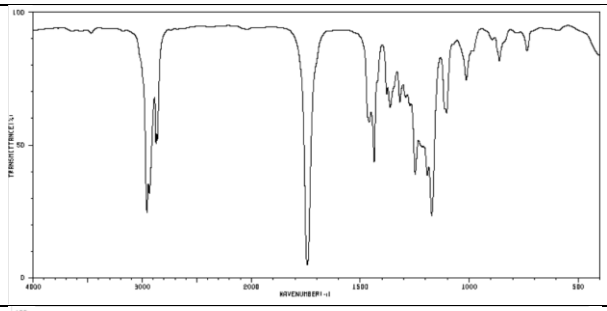
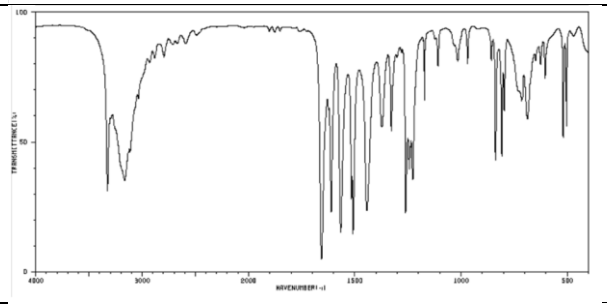
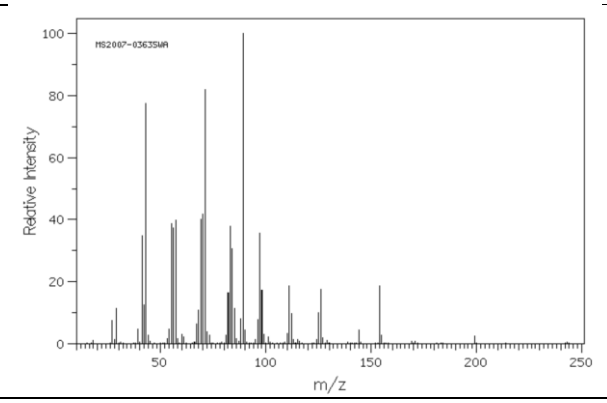
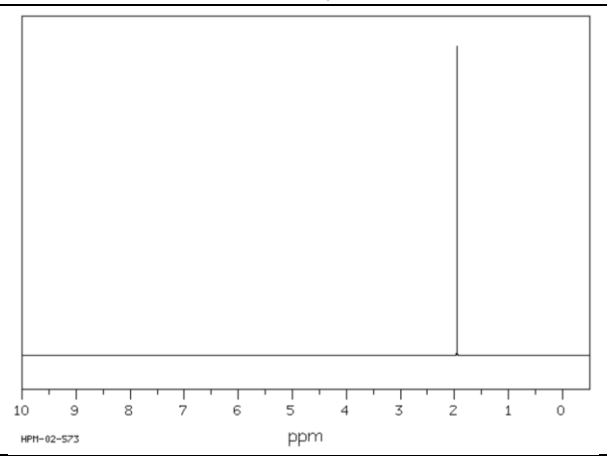
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Code	Spectrum
A	
B	
C	
D	

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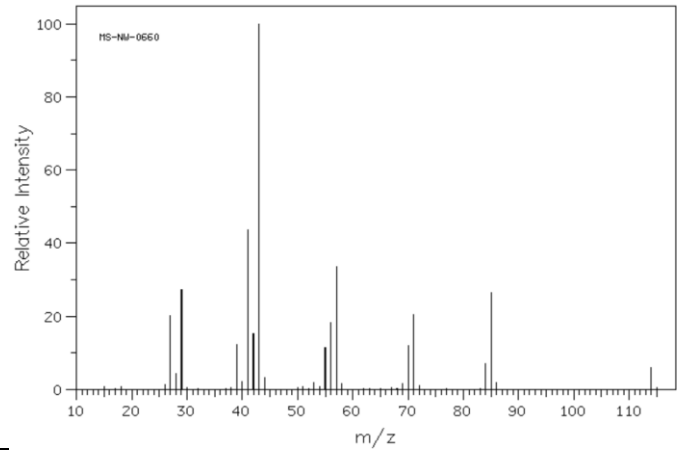
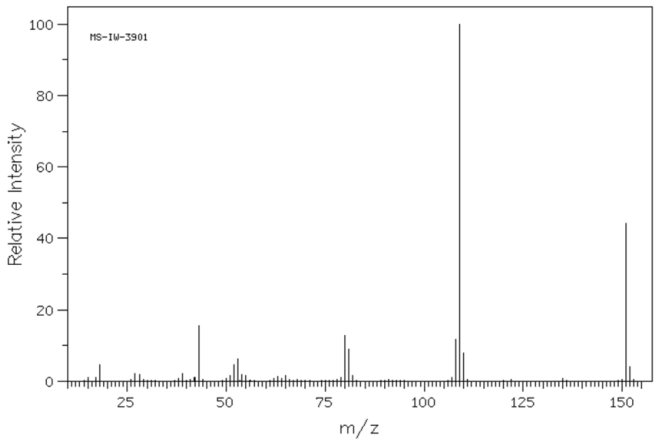
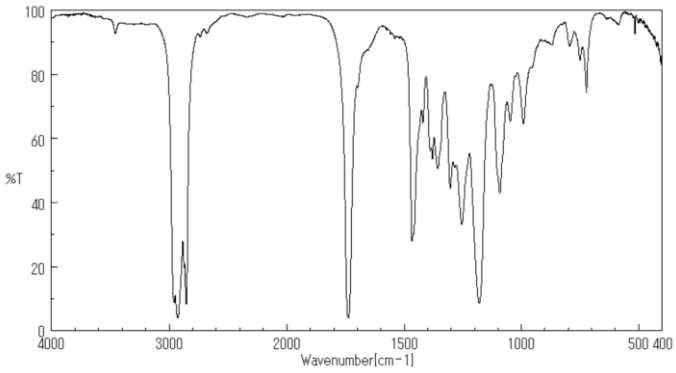
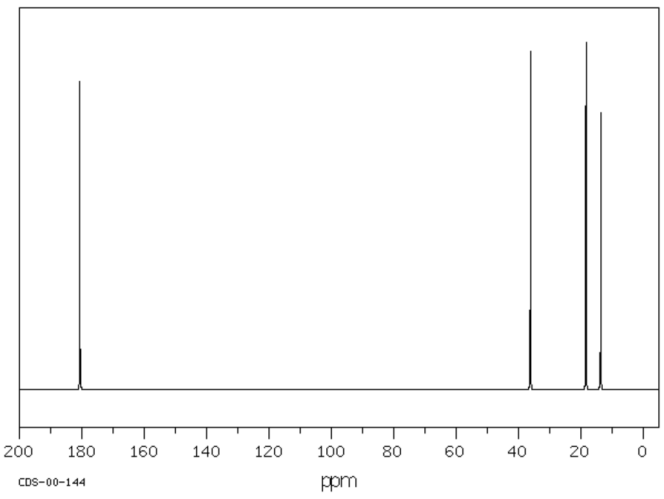
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E	
F	
G	
H	

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Downloaded by The-Bao Nguyen (ngu1628@stalbanssc.vic.edu.au)

I	
J	
K	
L	

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a. Identify the spectrum code (A-L) for the mass spectrum of **paracetamol**. Justify your response by referring to one piece of evidence on the spectrum.

2 marks

Spectrum code	
Justification	

b. Identify the spectrum code (A-L) for the infrared spectrum of **paracetamol**. Justify your response by referring to one piece of evidence on the spectrum.

2 marks

Spectrum code	
Justification	

c. Identify the spectrum code (A-L) for the ^{13}C NMR spectrum of **butanoic acid**. Justify your response by referring to one piece of evidence on the spectrum.

2 marks

Spectrum code	
Justification	

d. Identify the spectrum code (A-L) for the ^{1}H NMR spectrum of **butanoic acid**. Justify your response by referring to one piece of evidence on the spectrum.

2 marks

Spectrum code	
Justification	

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Question 10 (17 marks)

The concentration of ethanol in a reduced-alcohol white wine was determined using the following reaction

$2Cr_2O_7^{2-}(aq) + 3CH_3CH_2OH(aq) + 16H^+(aq) \rightarrow 4Cr^{3+}(aq) + 3CH_3COOH(aq) + 11H_2O(l)$

A 25.00 mL sample of the white wine was pipetted into a 250.0 mL volumetric flask and deionised water was added up to the calibration mark.

20.00 mL aliquots of the diluted wine were transferred into 5 conical flasks and titrated with 0.0502 M $K_2Cr_2O_7(aq)$.

The results are shown below

(mL)	**Titre 1**	**Titre 2**	**Titre 3**	**Titre 4**	**Titre 5**
Initial	5.0	5.05	27.25	7.25	24.15
Final	22.2	27.25	44.20	24.15	41.00
Titre	17.2	17.15	16.95	16.90	16.85

a. Define endpoint 1 mark

b. Circle any concordant titres 1 mark

c. Suggest an improvement to the recording of the value for the first titre 1 mark

d. Calculate the average concordant titre 1 mark

e. Calculate the concentration of ethanol in the diluted white wine sample 2 marks

f. Calculate the concentration of ethanol in the undiluted wine 1 mark

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g. The density of ethanol is 0.785 g mL^{-1} Calculate the concentration of ethanol in the original wine in % v/v. 3 marks

h. State and explain the impact of the calculated ethanol concentration in the wine if the endpoint was consistently overshot. 1 mark

i. The label declaration states the concentration of ethanol in the white wine is 5.5 % v/v. Discuss the accuracy and precision of the data obtained in the investigation

4 marks

	Definition	**Discussion**
Accuracy		
Precision		

j. Identify the dependent variable in this investigation 1 mark

k. Name the part of a scientific poster that should include the data analysis including explanation of exclusion of outliers 1 mark

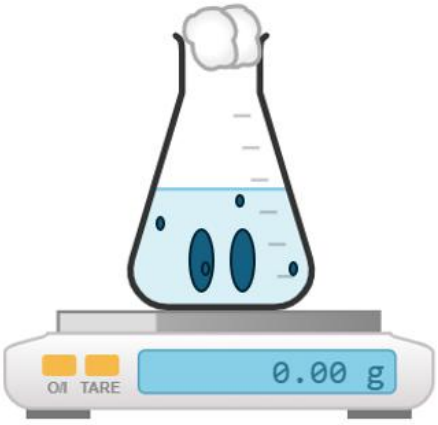
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Question 11. (11 marks)

Consider the results obtained from an experiment

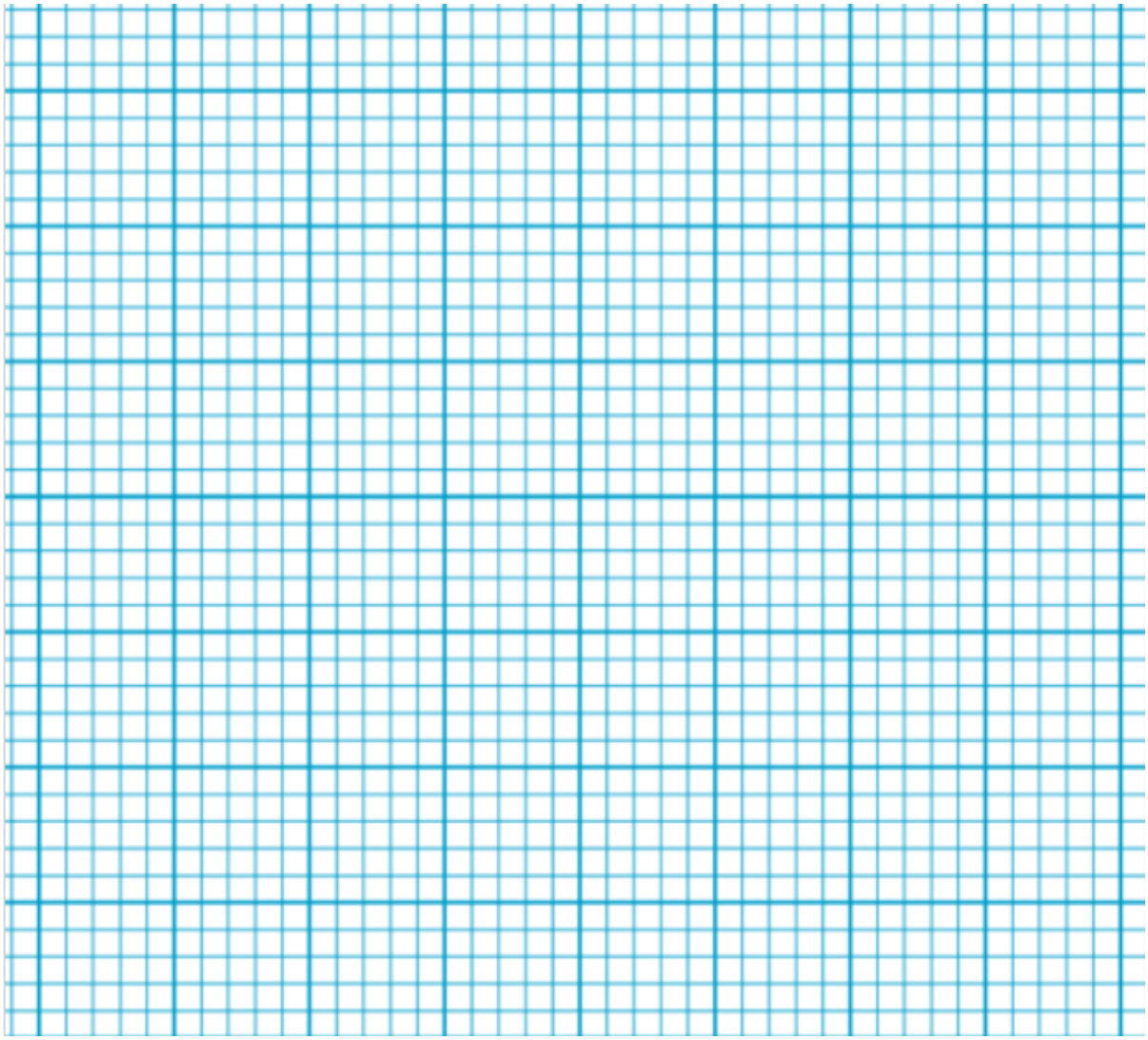


Time (s)	Mass of system (g)
0	33.02
1	32.77
2	32.46
3	32.23
4	32.01
5	31.91
10	31.41
15	31.12
20	31.09
25	31.07
30	31.05
35	31.05
40	31.05

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a. Plot a graph of the data and draw a line of best fit. Plot the range of mass of the system from 30.0 to 34.0 g (the scale does not need to go through 0.0).

2 marks

b. Circle the correct option for the trend shown 1 mark

Positive linear **Negative linear** **Positive non-linear** **Negative non-linear**

c. State the resolution of the recorded mass 1 mark

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d. Calculate the average rate over the period from 0 to 2s 1 mark

e. Calculate the average rate of reaction over the 40 s period 1 mark

f. Write a chemical equation for a reaction that could have been studied in this investigation 2 marks

g. Using appropriate chemical theory explain the trend observed in the graph 3 marks

END OF TRIAL EXAM

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